

SMART CAMPUS ENERGY MONITORING AND ENERGY-SAVING RECOMMENDATION SYSTEM USING C

1. TITLE

Smart Campus Energy Monitoring and Energy-Saving Recommendation System Using C

2. PROBLEM STATEMENT

An engineering college operates multiple facilities such as academic blocks, computer laboratories, libraries, hostels, canteens, administrative offices, and research laboratories, each consuming significant amounts of electrical energy on a daily basis. Due to increasing energy consumption and the need for sustainable campus management, the college management wants to monitor electricity usage and identify buildings with excessive or abnormal energy consumption.

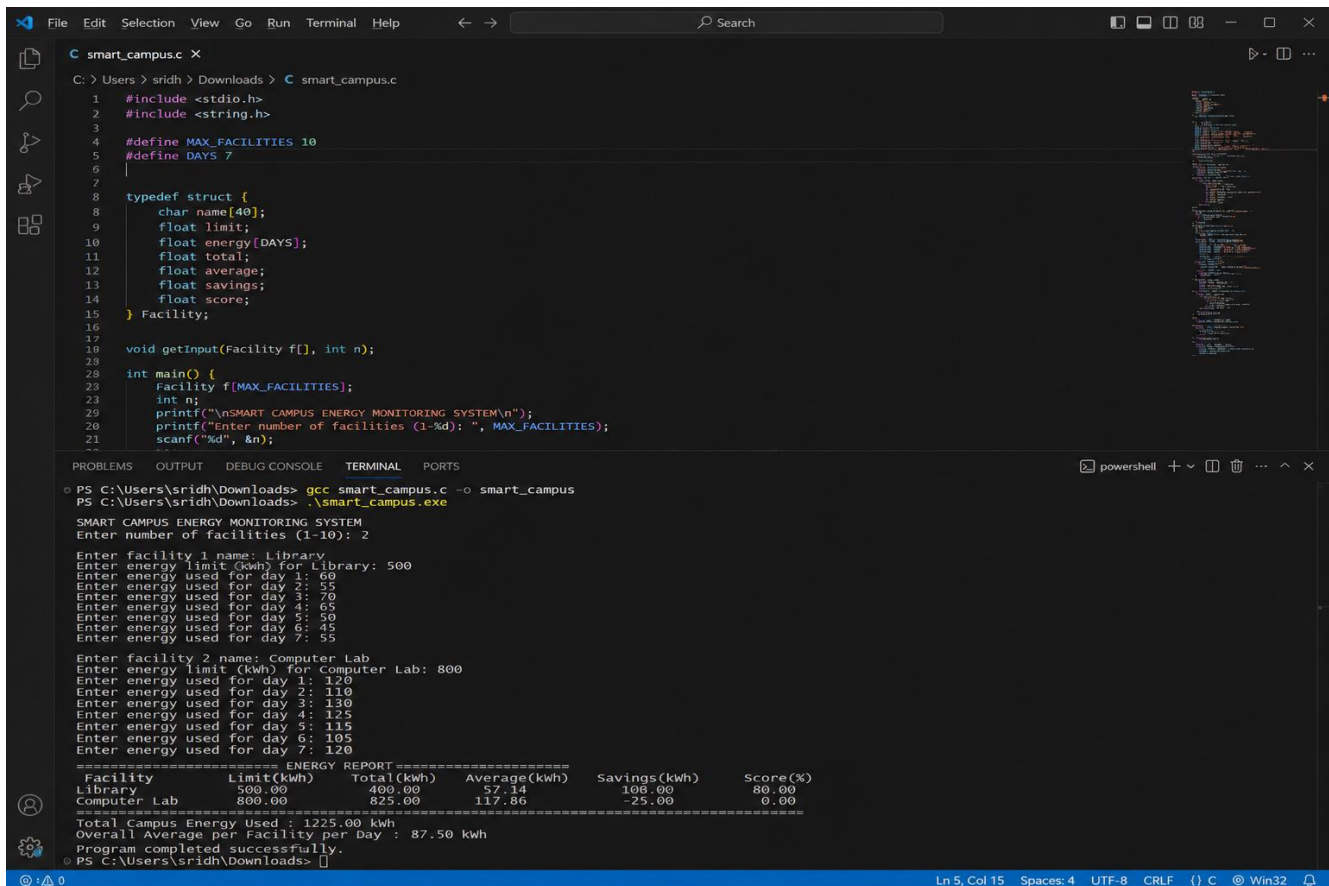
The system collects daily energy consumption data for different campus facilities over seven days and analyzes the data to determine total and average consumption, highest and lowest consumption days, facilities exceeding permitted energy limits, abnormal consumption patterns, and top energy-consuming facilities. It also calculates potential energy savings based on a reduction target, generates a sustainability score, and provides suitable energy-saving recommendations.

3. TEST CASE AND VALIDATION

Test Case	Input / Condition	Expected Result	Validation	Status
TC01	Valid facility count and positive 7-day values	Analysis calculated	Total and average displayed	Pass
TC02	Negative energy value	Value rejected	Program asks again	Pass
TC03	Average above permitted limit	Exceeds-limit status	EXCEEDS LIMIT displayed	Pass
TC04	Search existing facility	Record found	Name and average displayed	Pass
TC05	Search non-existing facility	Record not found	Not found displayed	Pass
TC06	Sort by consumption	Highest first	Sorted list displayed	Pass
TC07	Reduction target = 10%	Saving calculated	Potential saving displayed	Pass
TC08	Reduction target > 100%	Invalid target rejected	Validation message displayed	Pass
TC09	Save report	Report file created	energy_report.txt generated	Pass
TC10	Invalid menu choice	Invalid choice message	Program continues	Pass

4. RESULT (OUTPUT SCREENSHOT)

Run the separate smart_campus.c source file in VS Code and paste the actual program output screenshot below.



The screenshot displays the Visual Studio Code editor with the `smart_campus.c` file open. The code defines a structure for facilities, including fields for name, limit, energy usage over 7 days, total, average, savings, and score. It includes a `getInput` function and a `main` function that prompts the user for the number of facilities and their details.

The terminal output shows the execution of the program. It prompts the user to enter the number of facilities (1-10), and then for each facility, it prompts for the name, energy limit, and energy usage for each of the 7 days. The output for two facilities is shown: Library and Computer Lab.

ENERGY REPORT

Facility	Limit(kwh)	Total(kwh)	Average(kwh)	Savings(kwh)	Score(%)
Library	500.00	400.00	57.14	100.00	80.00
Computer Lab	800.00	825.00	117.86	-25.00	0.00

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Total Campus Energy Used : 1225.00 kwh
Overall Average per Facility per Day : 87.50 kwh
Program completed successfully.

The status bar at the bottom indicates the current line and column (Ln 5, Col 15), the number of spaces (4), the encoding (UTF-8), the line ending (CRLF), the file type (C), and the window state (Win32).